

Jaymin Sheladia

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EDUCATION

University of Southern California

Los Angeles, USA

Master of Science in Computer Science

January 2026 – December 2027

- **Coursework:** Analysis of Algorithms, Agentic AI, Applied NLP, Web Technologies, Database Systems

Graphic Era Deemed to be University

Dehradun, India

B.Tech in Computer Science and Engineering

October 2021 – June 2025

- **Coursework:** Operating Systems, Network & System Security, Computer Networks, OOPs, Software Engineering

WORK EXPERIENCE

Cognizant Technology Solutions

Coimbatore, India

Programmer Analyst Trainee – Software Engineering

May 2025 – September 2025

- Engineered a full-stack **travel planning and booking platform** using **Angular, Spring Boot, PostgreSQL, and Redis**, designing **15+ RESTful APIs** with JWT-based RBAC, transactional persistence, and caching.
- Designed an **event-driven booking architecture** using **Apache Kafka**, Redis caching, and Resilience4j circuit breakers, targeting **30–40% lower read latency** while maintaining idempotency under concurrent requests.
- Integrated an **LLM-assisted recommendation engine** combining user preferences, budget, and trip duration to generate personalized itineraries, containerizing services with **Docker** and targeting **80%+ test coverage**.

CODTECH IT Solutions

Remote

Machine Learning Intern

June 2024 – August 2024

- Engineered an end-to-end **fraud detection system** over **200,000+** credit card transactions, benchmarking XGBoost, Random Forest, Logistic Regression, and Isolation Forest under severe class imbalance.
- Optimized model features and decision thresholds to target **90%+ fraud recall**, productionizing the model through a **FastAPI** inference service with real-time risk scoring and batch inference.
- Developed a **model monitoring and retraining pipeline** tracking feature drift and prediction quality, targeting **under 100 ms** online scoring latency with Dockerized deployment and automated validation gates.

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, TypeScript, Python, SQL, HTML, CSS

Systems & Databases: PostgreSQL, MongoDB, MySQL, Redis, Database Internals, Query Optimization, Linux

Cloud & DevOps: AWS, Firebase, Docker, Kubernetes, Git, CI/CD, Microservices

Frameworks & Tools: React, Next.js, Node.js, FastAPI, Flask, Django, Express.js, REST APIs, GraphQL, GDB, GCC

AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, SpaCy, OpenAI APIs, Claude APIs

AI Dev Tools: GitHub Copilot, Cursor, Claude Code, ChatGPT, MCP, Claude/GPT APIs for agentic workflows

PROJECTS

TechHire GitHub | Python, FastAPI, LangGraph, React, PostgreSQL, pgvector, Redis June 2026 – September 2026

- Built a **RAG pipeline** over **1,040** job postings to improve search relevance, combining **pgvector embeddings** with **PostgreSQL full-text search** through **hybrid retrieval**, raising top-10 hit rate from **68%** to **92%**.
- Orchestrated **3 concurrent LLMs** with **LangGraph** into fault-tolerant summarization and citation-checked question-answering workflows, and added **Redis caching** that cut query latency **13.8x (19.1 ms to 1.38 ms)**.

Model Router GitHub | Python, FastAPI, SQLite

July 2026 – August 2026

- Architected a **policy-aware LLM routing service** layering identity verification, per-project authorization, and **reservation-based budget enforcement** across **Groq** models, with every decision logged to a SQLite audit trail.
- Built an **LLM-as-judge evaluation** benchmark against a hand-labeled oracle set, reaching **100%** classification accuracy at **33.2%** lower inference cost with **93.3%** quality retention vs. always calling the top-tier model.

Sequence Alignment: DP vs. D&C GitHub | Python, Algorithms

March 2026 – April 2026

- Implemented and comparatively evaluated classical **Needleman-Wunsch** dynamic programming against **Hirschberg's divide-and-conquer algorithm** for global DNA sequence alignment.
- Reduced memory usage from **$O(m \times n)$** to **$O(m + n)$** using Hirschberg's algorithm – up to **99.8%** lower peak memory on large inputs – verified against 200 trials to match optimal alignment cost.

RESEARCH WORK & PUBLICATIONS

Deep Learning Approach for Malicious URL Detection | Published Research Paper

2025

- Compared **4** deep learning architectures (**CNN, RNN, LSTM, Bi-LSTM**) for malicious URL detection, engineering character-level preprocessing pipelines on the IEEE Dataport dataset for sequential analysis at scale.
- Achieved **99.25%** accuracy with the Bi-LSTM model, outperforming CNN (**86.5%**), LSTM (**63.75%**), and RNN (**53%**) baselines across standard classification and error-rate metrics.